Problem-2

You're given an integer **N**. Write a program to calculate the sum of all the digits of **N**.

Input Format

The first line contains an integer **T**, the total number of testcases. Then follow **T** lines, each line contains an integer **N**.

Output Format

For each test case, calculate the sum of digits of **N**, and display it in a new line.

Constraints

- 1≤T≤10001≤T≤1000
- 1≤N≤10000001≤N≤1000000

Sample 1:

Input	Output
3 12345 31203 2123	15 9 8

Problem-3

Given an Integer **N**, write a program to reverse it.

Input

The first line contains an integer **T**, total number of testcases. Then follow **T** lines, each line contains an integer **N**.

Output

For each test case, display the reverse of the given number **N**, in a new line.

Constraints

- 1 ≤ **T** ≤ 1000

- $1 \leq \mathbf{N} \leq 1000000$

Sample 1:

Input	Output
4 12345 31203 2123 2300	54321 30213 3212 32

Problem-4

You are given an integer sequence $1,2,\ldots,n$. You have to divide it into two sets A and B in such a way that each element belongs to exactly one set and $|sum(A)-sum(B)|$ is minimum possible.

The value $|x|$ is the absolute value of x and $sum(S)$ is the sum of elements of the set S.

Input

The first line of the input contains one integer n

($1 \leq n \leq 2 \cdot 10^9$).

Output

Print one integer — the minimum possible value of $|sum(A)-sum(B)|$

if you divide the initial sequence $1,2,\ldots,n$ into two sets A and B

Examples

Input
3
Output
0
Input
6
Output
1